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VOLCANO

- Opening, or rupture, in a planet's surface or crust, allows to escape hot, molten rock, ash, and gases, from below the surface.

- **Occurrence** : Found where tectonic plates, pulled apart from each other or come together .

- Mid-oceanic ridge →Mid-Atlantic Ridge, examples of the volcanoes caused by "divergent tectonic plates" pulling apart.

  →The Pacific Ring of Fire has examples of volcanoes caused by "convergent tectonic plates" coming together.

- **Component of volcano :**

Liquid : Lavas
Solid : Tephra or Pyroclast.
Gas : 98% to 99% is steam, S, SO_2, CO_2, H_2S.

LAVA

- Underground molten materials, called **Magma**.
- When magma reaches on the ground, and air, called **Lava**.
- Occupied different shapes & types after cooling down.
- **Colors of Lava:** Black, red, gray, brown and tan, metallic sliver, pink, and green (due to presence of Olivine).



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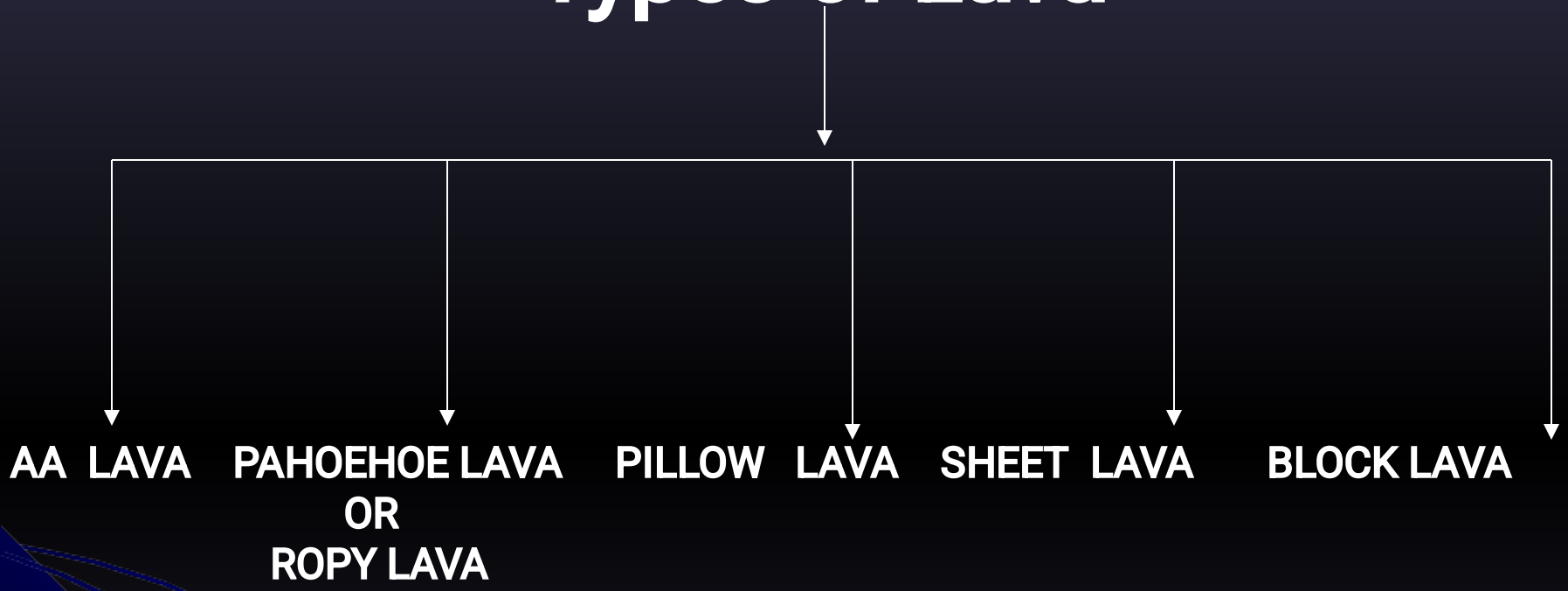


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Types of Lava





AA (Blocky Lava)



Pahoehoe (Ropy Lava)

Types of Lava

- **Aa Lava :**
- **Hawaiian** word, meaning "**stony with rough lava".**
- Having rough surfaces.
- Characteristic of viscous magmas.

- **Pahoehoe Lava :**
- **Hawaiian** word, meaning "**smooth, unbroken lava" .**
- Smooth surface, dense & form in large areas & contain little gas.
- If top surface, wrinkled & ropy, then it, called **Ropy lava**.

A-a lava (Ex : Glowing `a`a flow front advancing over pahoehoe on the coastal plain of Kilauea Volcano, Hawai`i.)



A-a Flow : Characters

- A`a flow
  `A`a (pronounced "ah-ah") is a Hawaiian term for lava flows that have a rough rubbly surface composed of broken lava blocks called clinkers. The incredibly spiny surface of a solidified `a`a flow makes walking very difficult and slow. The clinkery surface actually covers a massive dense core, which is the most active part of the flow. As pasty lava in the core travels downslope, the clinkers are carried along at the surface. At the leading edge of an `a`a flow, however, these cooled fragments tumble down the steep front and are buried by the advancing flow. This produces a layer of lava fragments both at the bottom and top of an `a`a flow.

Block Lava

- These lavas, which have surfaces covered by angular fragments, differ from aa in that the fragments have more regular forms and smoother faces. The surface blocks often approach cubes in form. Blocky lava flows form from more viscous lavas than aa flows, with the angular blocks formed by breaking up of the partly to wholly congealed upper part of the flow as still-mobile magma moves beneath the thick crust. These flows are typically thicker that aa lavas (8-35 m thick), and fragmental material, which may constitute the entire thickness, makes up a greater proportion of the flow than aa. The surfaces of blocky lavas are generally very irregular, with many hummocks and hollows, often 3-5 m deep.



Block Lava :

- Emerged in a fairly viscous state .
- Tend to be both stronger and thicker than Aa lava flows .
- Move slowly at a rate ranging from 1 to 5 meter a day.
- Characterized by often cubic masses with relatively smooth faces after solidification.
- Much less rough and pitted surface than Aa lava flows.

Pa-hoe-hoe lava (Ex.-Toes of a pahoehoe flow advance across a road in Kalapana on the east rift zone of Kilauea Volcano,Hawaii Hawai`i.)



- Pahoehoe
  Pahoehoe is a Hawaiian term for basaltic lava that has a smooth, hummocky, or ropy surface. A pahoehoe flow typically advances as a series of small lobes and toes that continually break out from a cooled crust. The surface texture of pahoehoe flows varies widely, displaying all kinds of bizarre shapes often referred to as lava sculpture


ROPY LAVA

Pae-hoe hoe Ropy lava




Pahoehoe Lava Flow
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PAHOEHOE LAVA



Pahoehoe lava flowing from a volcano in Hawaii.

Toothpaste lava

- Toothpaste lava, an important basalt structural type which illustrates the transition from pahoehoe to aa, is particularly well displayed on the 1960 Kapoho lava of Kilauea Volcano. Its transitional features stem from a viscosity higher than that of pahoehoe and a rate of flow slower than that of aa. Viscosity can be quantified by the limited settling of olivine phenocrysts and rate of flow by field observations related to the low-angle slope on which the lava flowed. Much can be learned about the viscosity, rheologic condition, and flow velocity of lavas long after solidification by analyses of their structural characteristics, and it is possible to make at least a semiquantitative assessment of the numerical values of these parameters.

Pillow structure

- **Pillow structure**

- Pillow lava are pillow-shaped structures sometimes seen in lavas and are attributed to the congealment of lava under water, or subaqeous extrusion. A pillow structure in certain extrusive igneous rock is characterized by discontinuous pillow-shaped masses, commonly up to 1 metre in diameter. Pillow lava commonly occuer at Constructive plate boundaries, forming part of a mid-ocean ridge. They are created when molten magma reaches the surface, but as there is a large difference in temperature between the lava and the water, the surface of the globule cools very fast, forming a skin. The skin cools a lot faster that the inside of the globule, so it is very fine grained, with a glassy texture. The magma inside the pillow cools more slowly, so is slightly coarser grained than the skin, but still classified as fine grained. There are often whole areas or lines of pillow lava. This is because when more magma rises from the mid-ocean ridge, the skin can no longer hold it in, so more pillow lavas erupt from the first one. Pillow lava can also formed when lava flows directly from the vent of the volcano into the sea.



- **Pillow Lava :**

- When lava poured under water, solidified rapidly & acquired characteristic Pillow structures.

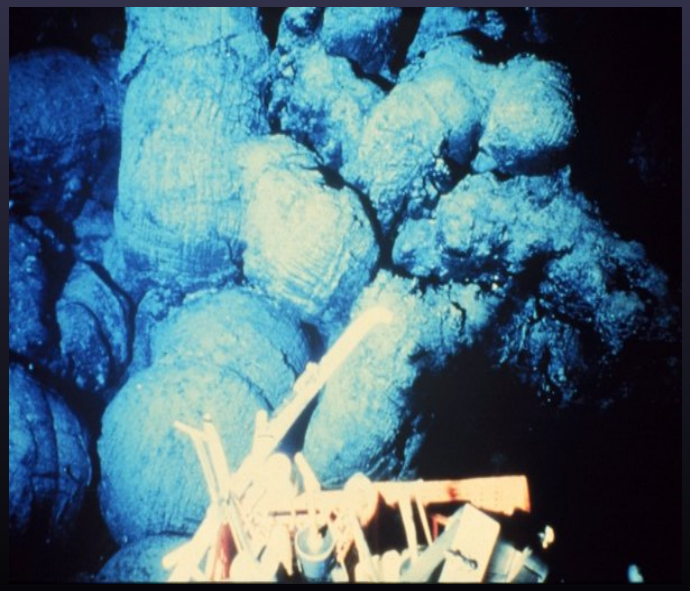
UNDER WATER
PILLOW LAVA





- Shape: Bulbous, spherical, or tubular lobes of lava.
- The marker of the younging direction.

PILLOW LAVA TUBES



FRESH PILLOW LAVA.



ORDOVICIAN PILLOW LAVA



PILLOW BASALT



- **Sheet Lava:**

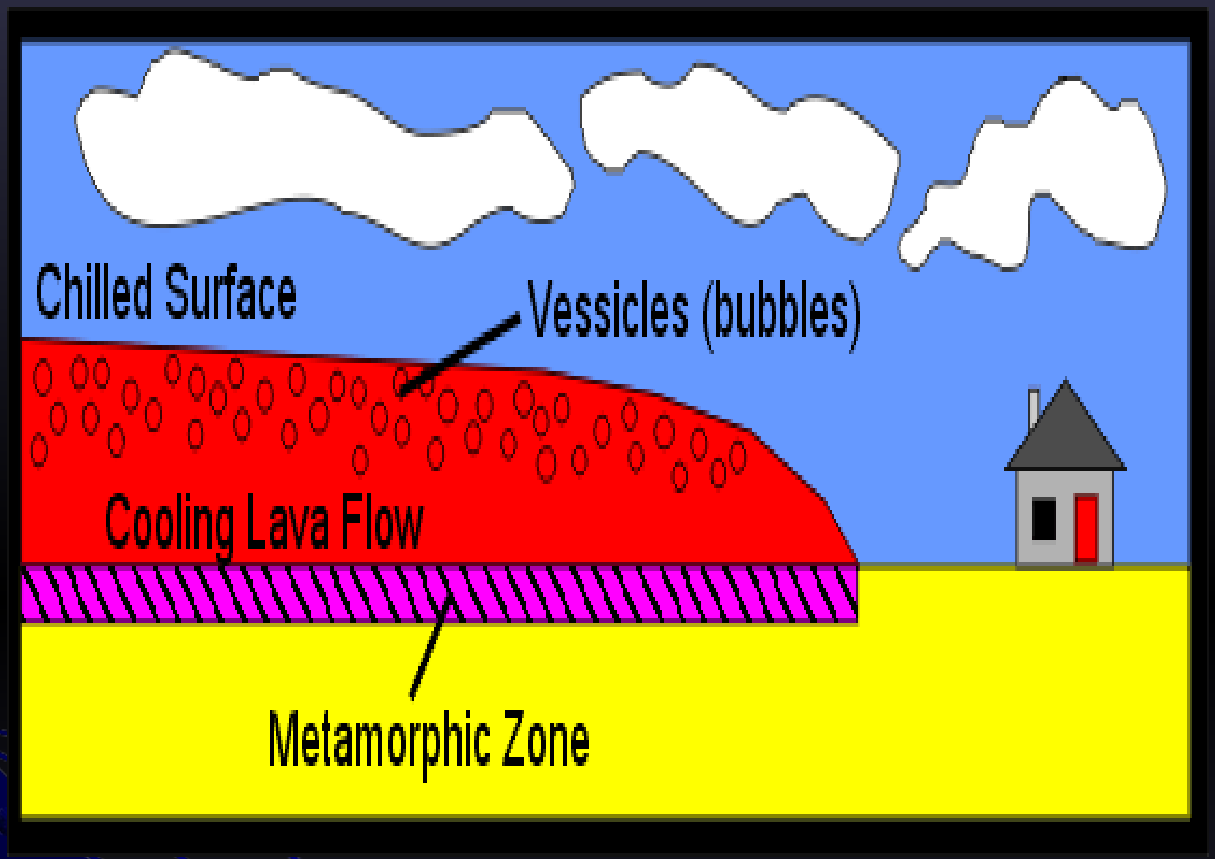
(Ex- Flood Basalt)

- Emerge from fissure systems, forming flows commonly ranging between 10m and 30m in thickness.
- Flow out so fast .
- Discharged over an enormous area.
- Ex- Deccan Flood Basalt



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A cooling lava flow metamorphoses underlying materials at it's base, and traps bubbles of gas beneath the rapidly chilled outer layer at the surface.

- **Vesicle :** The pores, created in the solidified lava after removal of gasses, called **Vesicles** .
- When vesicles, filled by the other minerals such as zeolite, chalcedony, quartz etc. then, seen like a nut, & called Amygdule & these structure, called **Amygdaloidal**.



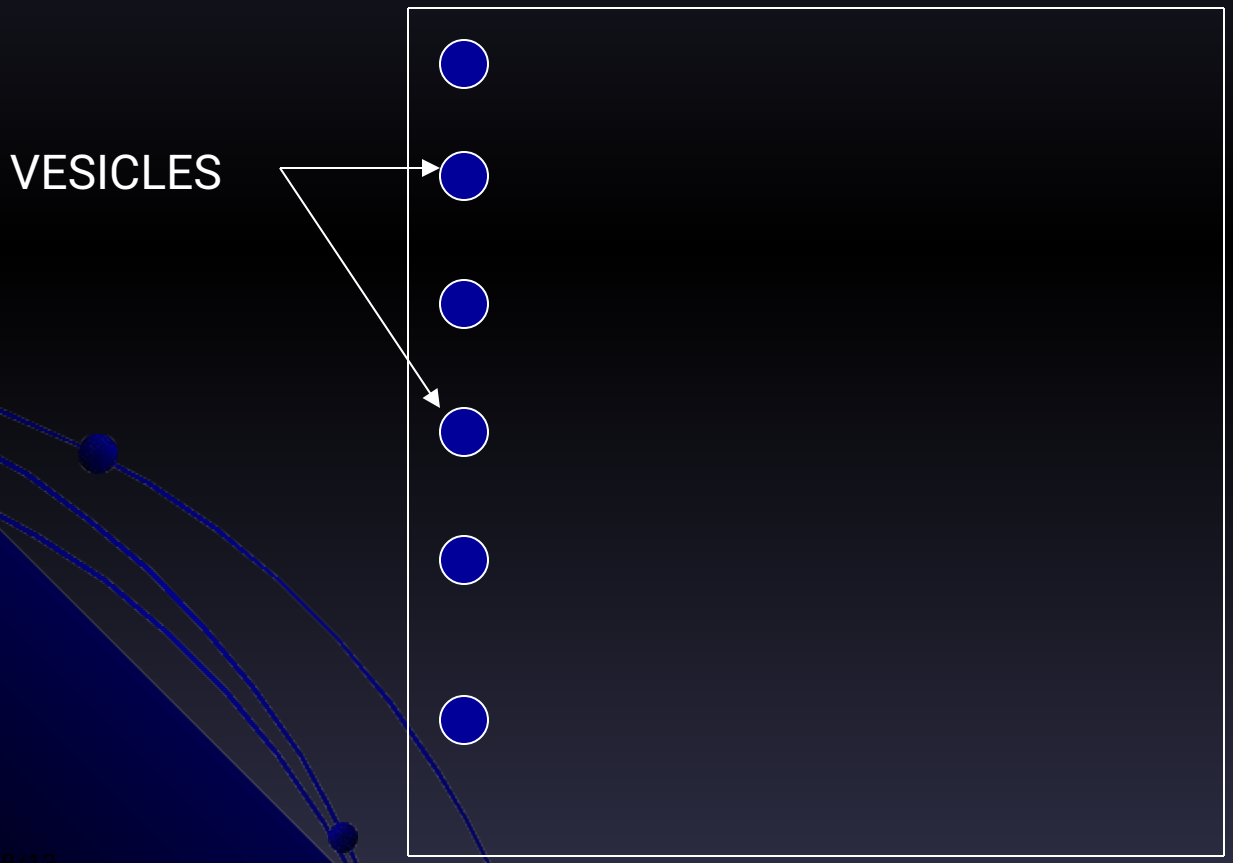
Basaltic boulders near coast. Criação Velha, Pico Island, Azores.



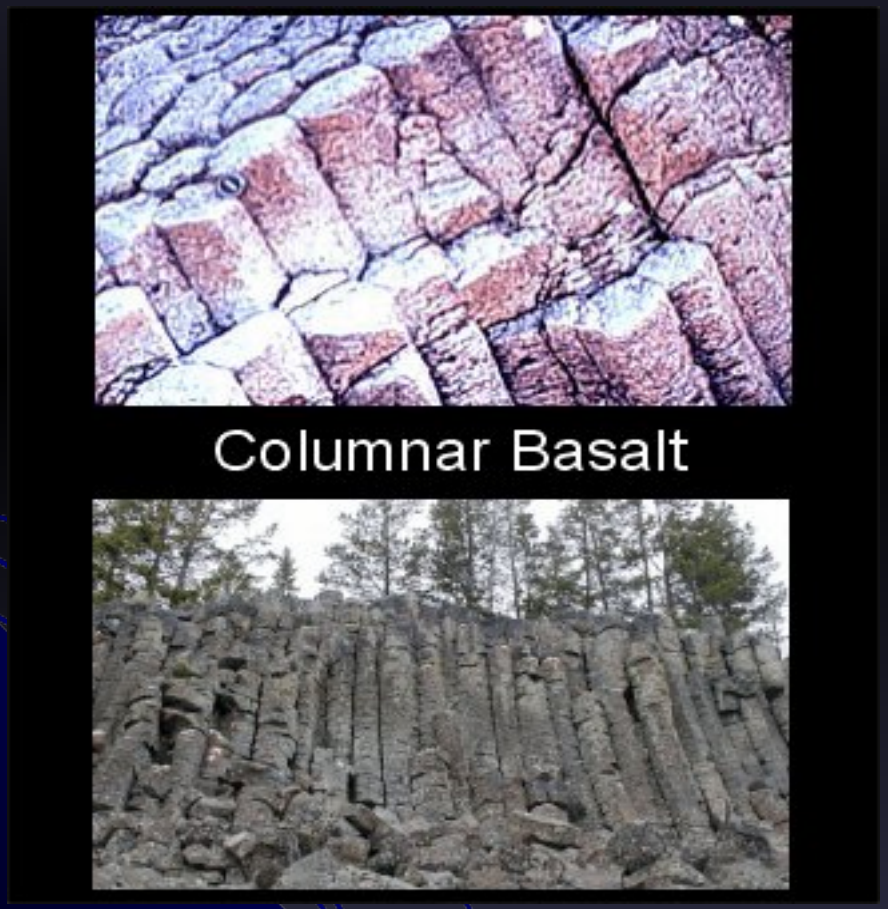
Piece of andesite lava (~4 inches across) newly formed near Brimstone Pit that shows elemental sulfur infilling the vesicles in the lava. [inset: Close-up of the sulfur which would have been liquid as it fills the cavities and has an origin in the magmatic gases sulfur dioxide (SO2) and/or hydrogen sulfide (H_2S). Image courtesy of Submarine Ring of Fire 2006 Exploration, NOAA Vents Program

Pipe Vesicles

When vesicles arrange as vertically, called **Pipe Vesicles.**



- # Columnar Basalt or Columnar Joint



Extensive, thick, homogeneous basaltic lava flows sometimes acquire hexagonal columnar structure, as they cool, called **Columnar Basalt** or **Columnar Joint .**

CULUMNAR BASALT



Skjalfandafljot, River, Iceland

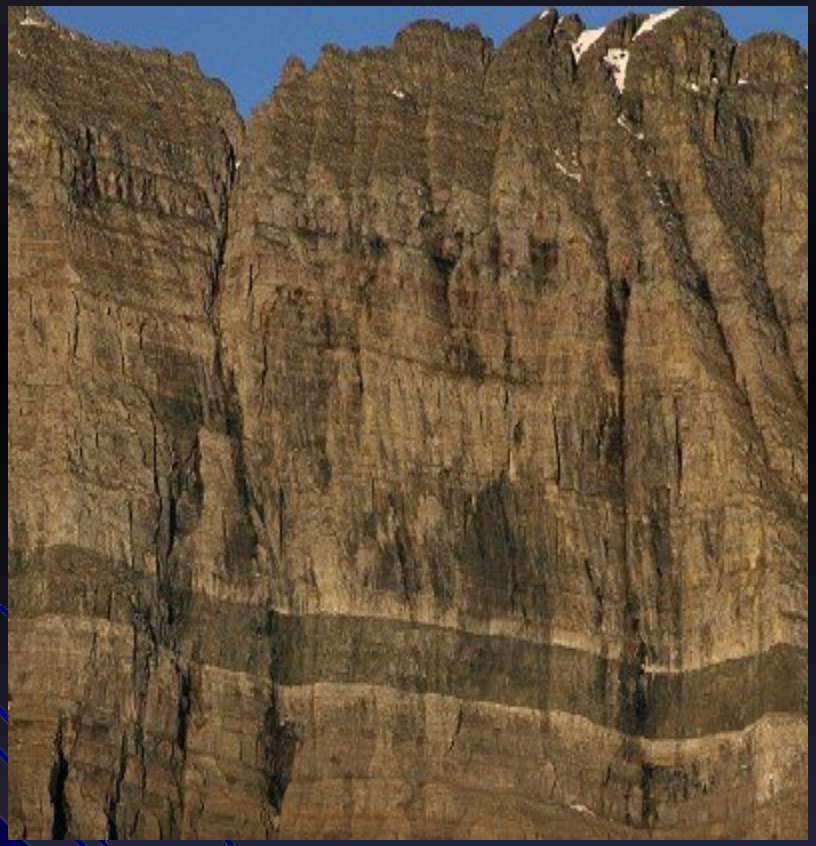

CULUMNAR
BASALT



A Tertiary igneous dyke, cross-cutting Permian wind blown sediments at Corrie shore.

- Dyke :

Vertical plutonic intrusion which x-cut other layers, called



- Sill :

Plutonic

What are Pyroclastics?

- **Volcanic bomb** : > 64 mm in diameter &
  circular or semicircular in shape.

- **Lapplli** : Ranging diameter → 64 to 2mm.

- **Ash**: Ranging diameter → less than 2 mm.



- When crater, filled by water, called **Caldera.**

Aniakchak Caldera formed during an enormous explosive eruption that expelled more than 50 km^3 of magma about 3,450 years ago. The caldera is 10 km in diameter and 500-1,000 m deep. Subsequent eruptions formed domes, cinder cones, and explosion pits on the caldera floor.

Types of Volcano :

1. Composite Volcanoes (also called **strato volcanoes**).
2. Shield Volcanoes (also called shields).
3. Cinder Cones.
4. Spatter Cones.
5. Complex Volcanoes (also called compound volcanoes).
6. Lava domes.
7. Super volcanoes.
8. Submarine volcanoes.
9. Sub glacial volcanoes.
10. Mud volcanoes.

COMPOSITE VOLCANOES OR STRATO VOLCANOES

- Tall conical mountains, composed of lava flows and alternate layers, therefore called Strato Volcano.
- Also known as Composite Volcanoes, created from several structures during different kinds of eruptions.
- Made up of cinders, ash and lava.
- Steeply dipping , dipping angle 30 to 50°.
- Examples include **Mt. Fuji in Japan**.

COMPOSITE VOLCANO



SHIELD VOLCANOES (also called shields)



Belknap Shield Volcano, Oregon

- Broad, shield-like profiles.
- Formed by the eruption of low-viscosity lavas.
- Flow a great distance from a vent.
- Not generally explode catastrophically.
- Very low slope, i.e. 0 to 5° .
- No pyroclast.
- Composed by central vent & subsidiary vents.
- E.g. **Mauna Loa and Kilauea (Hawaiian).**

• CINDER CONE



Hualalai Volcano, Hawaii. Photograph by J.P. Lockwood

- Steep conical hill .
- Result from eruptions .
- Lava fragments (i.e. pyroclasts & scoria), called **cinders**.
- Short-lived eruptions.
- Example : **Parícutin in Mexico**



SPATTER CONE



Lava fountain of the Pu`u `O`o cinder and spatter cone on Kilauea Volcano, Hawaii.

- Hot erupting lava contains just enough explosive gas to prevent the formation of a lava flow, but not enough to shatter it into small fragments the lava, torn by expanding gases into fluid hot clots, ranging in size from 1cm to 50cm across,

  called **Spatter**.



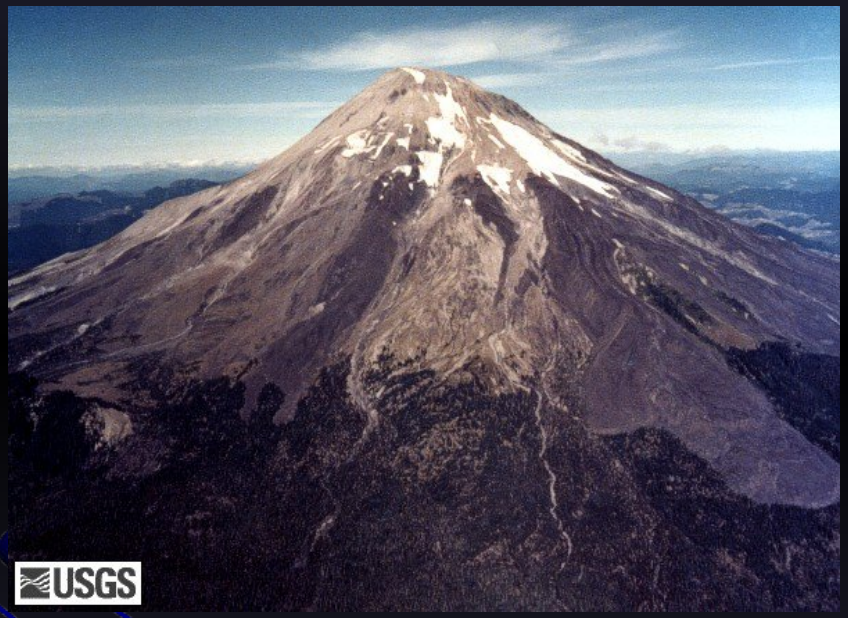
SPATTER CONE

Basaltic lava erupts from a vent on Kilauea Volcano, Hawai`i, to form row of fountains and lava flows. Basalt magma typically generates very little ash compared to other types (andesite, dacite, and rhyolite).

- When the **Spatter** falls back to Earth the clots weld themselves together and solidify forming steep-sided accumulations. These accumulations focused on an individual vent, called **Spatter Cones**.

• COMPLEX VOLCANOE

(also called compound volcano)



Mount St. Helens, Washington

- A volcano that consists of a complex or more vents, reckoned as a **compound** or **complex** volcano.

LAVA DOMES

- Built by slow eruptions of highly viscous lavas .
- Example : **Mount Saint Helens** .



LAVA DOME

The glowing and fast-growing lava dome of Kelud volcano in what is remaining of the crater lake.



LAVA DOME

Saint Helens, Mount: after eruption



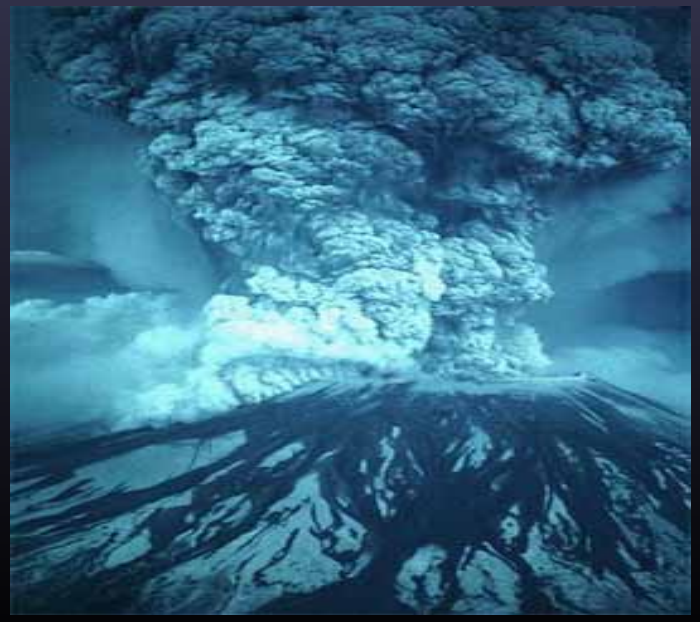
LAVA DOME

Mount St. Helens in Washington lava dome

SUPER VOLCANOES

- A large volcano, with a large **caldera.**
- Most dangerous type of volcano .
- Examples include **Yellowstone Caldera in Yellowstone National Park** and **Vales Caldera in New Mexico (both western United States), Lake-Taupo in New Zealand and Lake Toba in Sumatra, Indonesia.**
- Large igneous provinces, also considered as super volcanoes.

Yellowstone Super Volcano





SUBMARINE VOLCANOES

- Common features on the ocean floor .
- Pillow lava,common eruptive product of submarine volcanoes .


KAVACHI
SUBMARINE
VOLCANO

SUBGLACIAL VOLCAOES

- Sub glacial volcanoes develop underneath icecaps.
- Examples : **Iceland**.

SUBGLACIAL VOLCANO



Grimsvotn subglacial volcano on the Vatnajokull ice cap.

MUD VOLCANOES

- Largest structures, 10 km in diameter and reach 700 meters in height.

- **Lahars**, the type of mud flow. Pyroclast materials, mixed up with the rain water & comes down along the steeply slope.



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MUD VOLCANO

Mud volcano in the Norris Geyser Basin, Yellowstone National Park, Wyoming. The mud volcano is about 40 cm tall.


THANK
YOU